FINAL EXAM

Fall 2017

- This is a closed-book exam.
- Students whose native language is not English may use a dictionary if they wish.
- Please check that you have 6 pages in your copy of the exam.
- There are 180 points in the exam. You also have 180 minutes.
- Please put the first section, covering the shorter questions, in one book and the longer questions in a second book. Label all books.
- Remember to put your name on all books.
- Do not be concerned if your answers are not whole numbers.
- You cannot use your phone as a calculator. You may use your own calculator, but it cannot be web-enabled. Simple calculators will be provided by the MBA office, and are sufficient for completing all calculations in the exam.
- If your answer is a number put a box around it. Example: $\boxed{\text{P=\$436, Q=35,465 .}}$

Section I: Short Questions [64 points]

Please be concise. None of the answers require more than a brief paragraph.

1) [8 points] The Italian Government is considering imposing a specific tax of 10 Euros (per unit sold) on Luxottica, a monopoly seller of sunglasses and fashion eyewear. What will be the effect of this tax on i) prices, ii) quantities and iii) consumer welfare? Explain your reasoning for each clearly and be as precise as possible.

   Price will go up but not to the extent of 10 euros [3 points]
   Quantities will go down [3 points]
   Consumer welfare will go down [2 points]

2) [8 points] The supply of ocean-breeding fish like cod and bluefish is highly volatile from year to year depending on the ocean currents. Despite this volatility in supply, cod and bluefish prices are relatively stable. What does this tell you? Illustrate with a graph.

   If the demand curve is very flat, which seems likely given the substitution possibilities, movements in the supply curve do little to change prices.
   [Full credit once flat demand is mentioned. If someone talks about storing fish and ignores the demand curve give some points (2-5 points) for ingenuity.]

3) [8 points] The market for first-flush black Darjeeling tea is perfectly competitive. Your analysis of the financial statements of a large Darjeeling tea producer reveals that the firm is making millions of dollars of profits. Is this consistent with your understanding that long-run economic profits of a firm in a perfectly competitive market should be zero? Provide a precise explanation for your answer.

   [Full credit for one of the following explanations]

   (1) Economic profits can be zero while accounting profits can still be positive. Financial statements show accounting profits – so ignores opportunity costs.

   (2) When firms have economies of scale (e.g., when the ATC is U shaped), in <u>the short run</u> firms can earn a positive profit.

   (3) Alternatively, marginal firms in a competitive market makes zero economic profit, but this firm is more efficient than the marginal ones.

4) [8 points] A clawback provision is a special contractual clause sometimes included in employment contracts by financial firms, by which an executive may be required, under certain conditions, to pay back money already paid out. Explain precisely why you think such clauses exist. What specific economic problem do such provisions seek to address?

   Addresses moral hazard. [5 points]
   Seeks to prevent managers from taking excessive risks or engaging in fraud that may be difficult to observe in the short run, but become apparent in the long run. [3 points]

5) [8 points] You have been hired to provide research help to a government agency

investigating the pricing behavior of a monopolist. Your regression work indicates that at its current prices, the demand elasticity facing this monopolist is -0.5. Your boss looks at your results and tells you that she knows you are wrong. Why does she think so?

A monopolist never produces on the inelastic part of his or her demand curve. [5 points]
At an inelastic point the firm can increase price without losing many customers and thus increasing revenue. Or mentioning that the MR goes negative when the elasticity is < absolute value of 1, or being on the top half of a linear demand curve. [3 points]

6) [8 points] Is price elasticity of demand higher in the short run or the long run? Explain your answer with an examples.

Demand is less elastic in short run. [5 points]
Takes time to change behavior (example: Gasoline). [3 points]
[It is also ok if they say the opposite is true for some durable goods.]

7) [8 points] A new Yale startup will be producing custom-tailored suits. The suits can be manufactured with a mix of inputs: machines (m), and employee labor (l). After some experimenting, the managers have determined how effective each of the two inputs are, and have computed the marginal products of the two inputs: MP_m and MP_l. They also know the prices of the two inputs, p_m and p_l. In a cost-minimizing production plan, what will be the relationship between those four objects, and why?

Accept either $\frac{MP_l}{p_l} = \frac{MP_m}{p_m}$ or $\frac{p_m}{p_l} = \frac{MP_m}{MP_l}$ (make sure subscripts are correct). [4 points]
When this relationship holds, the firm has equalized the "return on investment" or "bang for the buck" from its purchase of inputs. Anything along those lines is fine. [4 points]

8) [8 points] Publishers usually first sell the hardbound edition of a book, followed 12 to 18 months later with a the paperback edition of the same book. What do you think is the reason for this widespread practice? Why in your opinion do the two editions not come out at the same time?

This is intertemporal price discrimination (a special case of 2^{nd} degree price discrimination) or self-selection. Publisher wants to separate two groups: the ones who are willing to spend money to read the book immediately and get the hardbound edition for their libraries, and the ones who just want to read the book (more price sensitive). [6 points]
Without an interval, some from the first group may be tempted to buy a paperback edition and save some dollars. [2 points]

Section II: Long Questions – Total of 116 points

1. [20 points] World Copper Market

You have been hired by the American Bureau of Metal Statistics to analyze various metal markets. You find that the demand for copper in 2006 was given by Q_d =36-6P, where Q is the quantity of copper in million metric tons per year, and P is the price in dollars per pound. The industry supply curve in 2006 was given Q_s = -12+18P.

a) [4 points] What was the equilibrium price and quantity of copper?

-12+18P=36-6P implies P= 2, Q=24 [2 points for P, and 2 points for Q]

b) [3 points] In 2007 because of the financial crisis, the demand for copper fell by 10%. (In other words, the demand in 2007 is Qd^{new}=0.9 Q_d.) As a consequence, the market price fell as well. Can you predict the percentage change in price caused by this 10% drop in demand?

-12+18P=0.9(36-6P)
i.e. -12+18P=32.4-5.4P
implies P= 1.897, Q=22.15 [2 points for the right P and Q]
So percentage change in price is -5%. [1 point for the percentage change]

c) [3 points] An intern at the American Bureau of Metal Statistics suggests that this change in demand and price can be used to quickly compute the price elasticity of the demand for copper. Do you agree with this view? Explain clearly why or why not.

No, this is incorrect. The demand curve itself has moved – so you cannot use this percentage change to compute elasticity. To compute elasticity you have to think about how price changes along the same demand curve affect demand.

d) [4 points] Compute the price elasticity of demand at the market equilibrium price in 2006.

Elasticity=(dQ/dP)*(P/Q)=-6*(2/24)=-0.5
[2 points for the correct formula if the final answer is wrong]

e) [3 points] Due to widespread strikes in the copper mines in Chile in 2008, the world supply of copper fell by 5%. What happened to equilibrium price and quantities of copper? (Assume that the demand was the same as in 2007.)

0.95*(-12+18P)=0.9(36-6P)
implies 22.95P=43.8
implies P=1.95 [2 points]
and Q=21.88 [1 point]

f) [3 points] Suppose that these same strikes had happened in 2006 instead. Can you compare qualitatively the relative effects on price and quantity that you would have seen compared to what you see in 2008? (***You do not need to re-compute the new price and quantity in 2006 with the strikes. A qualitative answer will suffice.)***

If you compare the demand curves of 2006 and 2008, the 2006 demand is less steep. This means, the same supply shift would have had a relatively higher effect on quantity than price in 2006.

2. [20 points] Basics of game theory

Consider the following simultaneous-move game:

Player 2

Player 1		L	C	R
	T	9, 5	5, 6	2, 4
	M	6, 5	2, 4	1 ,3
	B	2, 7	3, 6	4, 8

a) [6 points] Does player 1 have a dominated strategy? If so, please specify. Does player 2 have a dominant strategy? If so, please specify.

Yes, player 1 has a dominated strategy M. [3 points]

No, player 2 doesn't have any dominant strategy. [3 points]

b) [6 points] What are the Nash equilibria of this game?

The game has two NE: (T, C) and (B, R) [3 points for each]

c) [8 points] Suppose now that it becomes a sequential-move game and player 1 moves first. What is your prediction of the outcome?

Player 1 will now choose M, and player 2 will then choose L. [4 points for each. It is fine if they do not draw the game tree.]

3. [18 points] Monorail Services from Evans Hall to NYC

New Haven Monorail is a monopoly provider of monorail services that will run from Evans Hall to Grand Central Terminal in NYC. The demand for this monorail is given by Q=500-2P, or equivalently, P=250-Q, where Q is the number of people on each trip, and P is the price in dollars. For each trip, New Haven Monorail has fixed costs of $2000 plus a per person cost of $50 per passenger. (You can round up your answer for the number of people.)

a) [3 points] How many people will be on each trip?

P=250-Q/2
MR=250-Q
Setting MR=MC gives 250-Q=50 implies Q=200.
[2 points if MR is right but the final answer is wrong.]

b) [3 points] What is the profit-maximizing price that New Haven Monorail will charge?

P=250-100=150.

c) [2 points] What will be the profits in each trip?

Profit= 200*(150-50)-2000=18000

d) [10 points] It is well-known that there are two distinct segments of customers: Yale students and Yale faculty. Student demand is given by Q^S=200-1.2P while the faculty demand is given by Q^S=300-0.8P. Suggest a pricing strategy that will enable New Haven Monorail to earn more profits than it earned in part c) above. Be precise and quantitative in your answer. Indicate how much demand you expect from each segment and total profits for New Haven Monorail.

Student segment:
P=(200/1.2)-Q/1.2
Setting MR=MC gives
(200/1.2)-2*Q/1.2=50

200-2Q=60

implies Q=70 and P=108.33

[2 points for the right MR, and 2 points for the right Q]

Faculty segment:
P=(300/0.8)-Q/0.8

Setting MR=MC gives
(300/0.8)-2*Q/0.8=50

300-2Q=40

imples Q=130.

Implies P=212.50.

[2 points for the right MR, and 2 points for right Q]

Total profits= (108.33-50)*70+(212.50-50)*130-2000 = 232087.1 [2 points]

4. [15 points] Market for Shale Oil

In mid-2014, the (short-run) price of oil hovered around $115 and the US saw a boom of "fracking", which is a method to produce oil from shale formations. As a result, many identical firms entered. The annual total cost function for a typical shale oil firm, in thousands of dollars, as a function of the number of barrels of oil produced (in thousands) is:

$$TC(q) = 180 + 45q + 0.2q^2$$

Fixed costs represent the costs of drilling wells and land leases, and are due **at the beginning of every year**.

a) [3 points] Solve for the firm's marginal cost, average total cost and supply functions:

[1 point] To get the marginal cost, take the derivate of total cost with respect to q:

$$mc = \frac{\partial TC}{\partial q} = 45 + .4q$$

[1 point] Average total cost is given by:

$$ac = \frac{TC}{q} = \frac{180}{q} + 45 + .2q$$

[1 point] As the firms take the price of oil as given, the supply function can be found by equating price and marginal cost, then solving for q:

$$p = 45 + 0.4q_s$$

$$\Longrightarrow q_s = (p - 45)/0.4$$

[Full credit even if they don't invert to get in terms of q; e.g. inverse supply is fine]

b) [4 points] What is a typical firm's output and profit?

In the short run, the current prevailing wholesale market price is $115 per barrel.

Plug in the market price into the supply function, yielding:

$$q = 175 \text{ [2 points]}$$

With this, we can solve for a firm's total profit:

$$\pi = TR - TC$$

$$= (115 * 175) - 180 - 45(175) - 0.2(175^2) = \$5{,}945$$

Given the units are in 1,000s, the total profit is almost 6 M in profit. [2 points]

c) [4 points] Shale deposits have been discovered worldwide, although the US is by far the leader in shale oil production. If there were truly free entry and no constraint on the number of firms that could enter worldwide, what would we expect the long-run price of shale oil to become?

Approach 1: We know that in the presence of free entry that in the long-run firms will make zero economic profit. This implies that the long-run price of oil will be the price such that the average total cost is at its minimum.

$$\frac{\partial ATC}{\partial q} = -\frac{180}{q^2} + .2$$

Setting this equation equal to zero and solving for q, yields:

$$q^2 = 900 \;\Rightarrow q = 30 \text{ [3 points]}$$

The long-run price can then be found by evaluating the average total cost function at q = 30:

$$ATC(30) = 57 \text{ [1 point]}$$

Approach 2: You could set MC=AC to get min AC level.

$$\frac{180}{q} + 45 + .2q = 45 + .4q$$

which would give

$$\frac{180}{q} = .2q \text{ or } q = 30 \text{ [3 points]}$$

The long-run price can then be found by plugging in q = 30 in ATC function:

$$ATC(30) = 57 \text{ [1 point]}$$

d) [4 points] However, shale oil competes with oil produced from traditional wells, which tend to have far lower costs. The price of oil fell greatly in 2015 as total output expanded. In the short run, how much oil will a typical firm produce if the price falls to $50 per barrel and what will their profits be? How much oil will they choose to produce the following year if they expect the price to remain at that level?

Recall that the supply function that you solved for in part (a) is given by

$$q_s = (p - 45)/0.4$$

Evaluating this function at p = 50, yields:

$$q_s = 12.5 \text{ [2 points]}$$

For the second question [2 points]:

Approach 1: Price 50 is below the min ATC, so the firm should exit and produce nothting

Approach 2: Calculate the profit

$$\pi = TR - TC$$

$$= (50 * 12.5) - 180 - 45(12.5) - 0.2(12.5^2) = -148.75$$

Each firm will thus lose a total of $148,750 and will exit the following year.

5. [23 points] Market for insurance

Consider the following simple model of health insurance. Let there be three types of people, LOW, MEDIUM, and HIGH risk, whose yearly medical expenses are $0 with 95% probability and $25K, $50K, and $100K (respectively) with 5% probability for the LOW, MEDIUM and HIGH groups. All three risk types are equally represented in the population, and all people have an initial wealth level of $100K, and a utility function over wealth of $u(x) = \sqrt{x}$ (where x is measured in thousands of dollars).

Insurers only offer the simplest policy, where people pay a premium but no deductible and medical expenditure is fully reimbursed. Assume that insurers have no costs other than the expected payouts to their policyholders.

a) [9 points] Compute the expected cost to the insurer for each type of customer. Also, compute the highest premium that each type of consumer is willing to pay: You can simply copy the following table and fill it out.

Customer risk type	Expected cost of customer type to insurer	Highest premium consumer would pay
LOW		
MEDIUM		
HIGH		

The highest premium a person is willing to pay, p (where l is the loss in the bad state of the world) is the solution to $5\% \times \sqrt{100-l} + 95\% \times \sqrt{100} = \sqrt{100-p}$. Expected cost to the insurer is $5\% \times l$. Filling out the table:

Customer risk type	Expected cost to insurer	Highest premium consumer would pay
LOW	$1250	$1335
MEDIUM	$2500	$2907
HIGH	$5000	$9750

[1 point for each number in the cost column; 2 points for each number in the premium column]

b) [4 points] If it were required that all people be insured, and insurance is competitive so that premiums equal average cost, what would the premium be?

The average cost is (1/3)*($1250+$2500+$5000)=$2917. This is the actuarially fair premium.

c) [10 points] If insurance were optional, and again insurance is competitive so that premiums equal average cost, what premium would insurers charge? And which risk types would choose to buy insurance at that premium? (Hint: Note that they would consider charging one of the values solved for in part a. If they charged the lowest, all three types would buy; if they charge the 2nd lowest, two types would buy; if they charged the highest, only one type would buy).

If the insurer prices at $1335 or below, its average cost will be (1/3)*($1250+$2500+$5000)=$2917. Negative profit.

If the insurer prices between $1336 and $2907, its average cost will be (1/2)*($2500+$5000)=$3750. Negative profit.

If the insurer prices between $2908 and $9750, its average cost will be $5000.

So it can avoid negative profit if it serves HIGH types only, at an actuarially fair premium of $5000.

If the student only says that only HIGH types will be served at a price of $5000, with no more explanation, 5 points.

If they explain why the other two prices won't work, full credit.
If they are able to rule out one of the incorrect options (but don't get the full solution), give 3 points partial credit.

6. [20 points] A game of add-on pricing

Consider a price competition between two movie theaters. They sell both movie tickets and popcorn. Suppose they are offering the same film and the same type of popcorn. The cost of each movie ticket is $20, and consumers' willingness to pay is $30. The cost of each tub of popcorn is $2 and consumers' willingness to pay is $8. Suppose the movie ticket price is publically observable but the popcorn price is not (which is usually the case in the market). As a result, when consumers choose which theater to go, they compare the ticket price only and always choose the cheaper one, but forget to take into account that they will also buy popcorn once they arrive at a theater as long as its price does not exceed their willingness to pay. We call such consumers "naïve consumers". Suppose each consumer will buy at most one tub of popcorn. Suppose the number of consumers is not big, so there is no need to worry about capacity constraint. Suppose the two movie theaters choose their prices for the movie ticket and popcorn simultaneously (and only once). If a consumer is indifferent between the two movie theaters, she will randomly pick one.

a) [8 points] What is the price of popcorn in the Nash equilibrium? What is the price of the movie ticket in the Nash equilibrium? Explain your answers.?

Since the price of popcorn does not affect consumers' choice of where to go, popcorn will be sold at the monopoly price $8. [4 points]
Ticket will be sold at a subsidized price $14 (=20-6), where 6 is the profit from selling the popcorn. That is, the profit from popcorn will be competed away in attracting consumers by setting a low ticket price. [4 points]

b) [7 points] Suppose now that all the consumers become "savvy": they find out the popcorn prices and take them into account when they choose which movie theater to go. Then what prices will the movie theaters set for their tickets and popcorn, respectively, in the Nash equilibrium? What about if only half of the consumers become "savvy"?

If both prices are observable, one NE is $2 for popcorn and $20 for the ticket like in the standard Bertrand competition. [4 points. There are many other NE: any price combination with $x+$y=$22 is a NE. Give full credits for any right NE.]

If only half of the consumers are savvy, then NE is $8 for popcorn and $14 for the ticket, the same as in question a). [3 points]
[Here is the reason (but this is not needed for full credits): Savvy consumers will compare the bundle price, but the theaters are already making zero profit, so they cannot offer better prices.

There is no other NE. E.g., ($2,$20) is not a NE, because otherwise each theater would have a unilateral incentive to set a lower ticket price, say, $19 to attract naïve consumers and in the same time make money from charging a higher popcorn price, say, $5. Losing savvy consumers does not matter because no profit from them anyway.]

c) [5 points] Return to the situation where all consumers are naïve. Suppose the city now tries to protect consumers by setting a price cap $6 for popcorn. Will consumers benefit from this intervention? Please explain your answer.

No, consumers won't benefit from this price cap. [3 points]
With the price cap, popcorn will be sold at $6, but then the movie ticket price will rise to $16 (=20-4) since now attracting a consumer is less profitable than before. So consumers pay the same bundle price $22 as before. [2 points]